

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:23

Aim

To sharpen an image using the Unsharp Masking technique.

Principle

Unsharp masking sharpens an image by subtracting a blurred version of the image from the original image.

Formula: where:

- = Original image
- = Blurred image
- = Sharpened image

Note: The formula in your provided image gives the high-frequency/detail mask. In practical image sharpening, the final sharpened image is usually obtained by adding this mask back to the original:

Sharpened Image = Original Image + Mask

Algorithm

1. Read the input image.
2. Convert the image to grayscale.
3. Apply Gaussian Blur to obtain the blurred image.
4. Subtract the blurred image from the original image to obtain the unsharp mask.
5. Add the mask to the original image.
6. Display the original, blurred, mask, and sharpened images.

Code:

```
EXP-23.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure BEN 10.png is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Blur the image
blurred = cv2.GaussianBlur(gray, (5, 5), 0)

# Step 4: Create the unsharp mask
mask = cv2.subtract(gray, blurred)

# Step 5: Create the sharpened image
sharpened = cv2.add(gray, mask)

# Step 6: Display the images
cv2.imshow("Original Image", gray)
cv2.imshow("Blurred Image", blurred)
cv2.imshow("Unsharp Mask", mask)
cv2.imshow("Sharpened Image", sharpened)

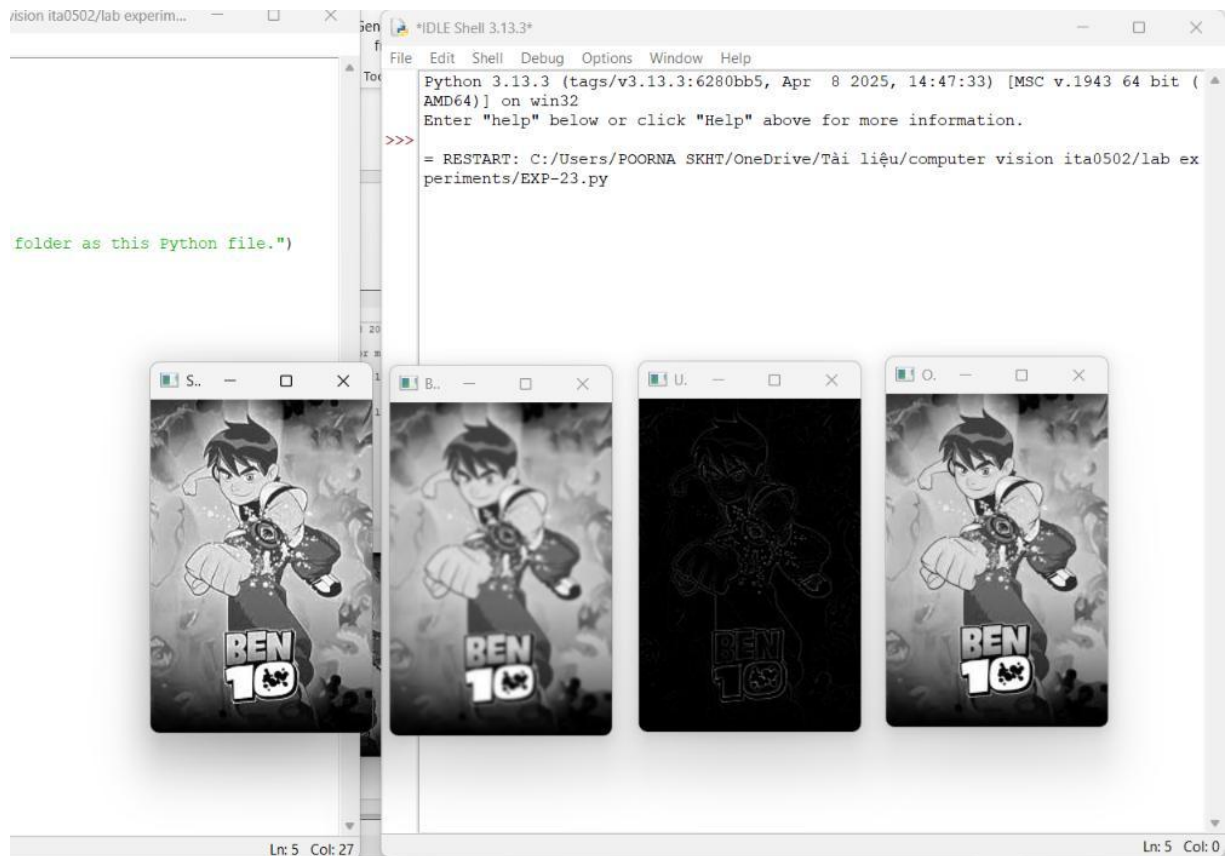
# Step 7: Wait for a key press
cv2.waitKey(0)

# Step 8: Close all windows
cv2.destroyAllWindows()

Ln: 5 Col: 27
```



OUTPUT :



Result

The image is successfully sharpened using the Unsharp Masking technique. The blurred image is subtracted from the original to extract high-frequency details, and these details are added back to produce a sharper image.